

Week One Project Plan

AI Based Biodiversity Monitoring Using Video Analysis

Week Goal	Establish and evaluate the initial AI wildlife detection system
Main Focus	Pretrained YOLO model, wildlife data and video detection
Expected Result	A working prototype that detects selected animals and humans from video

Day One

- Set up the Python environment
- Install and configure the YOLO model
- Load a pretrained YOLO model
- Collect a few wildlife images and videos
- Run initial object detection tests
- Save sample detection results

Expected result A basic YOLO detection test working on wildlife footage

Day Two

- Research suitable wildlife datasets
- Shortlist the target animal species
- Select approximately five to eight species
- Organize the initial dataset
- Check the quality and availability of images

Expected result Final target species and an initial dataset

Day Three

- Test YOLO on different wildlife images and videos
- Observe detection confidence
- Identify false detections
- Identify species that require better training data
- Record initial observations

Expected result Initial detection performance observations

Day Four

- Evaluate whether the pretrained model is sufficient
- Compare results across selected species

- Decide whether custom fine tuning is required
- Finalize the AI approach for the project

Expected result A clear decision between pretrained use and fine tuning

Day Five

- If required prepare the selected dataset for fine tuning
- Create training and validation data
- Start basic model improvement
- If fine tuning is not required optimize the existing detection settings
- Test different confidence thresholds

Expected result Improved detection performance and a stable AI configuration

Day Six

- Build the basic video detection pipeline
- Use OpenCV to read video frames
- Pass frames through YOLO
- Display detected animals and humans
- Generate a processed result video

Expected result A complete video detection pipeline

Day Seven

- Test the complete Week One prototype
- Review detection results
- Document the selected species
- Document model observations
- Prepare a short progress demonstration for the supervisor

Expected result A working initial wildlife detection prototype and Week One progress report

Week One Final Pipeline

Wildlife Video
OpenCV
YOLO Detection
Animal and Human Detection
Detection Results

Important scope for Week One Focus only on the AI detection component. Backend, database, dashboard, audio recognition and advanced tracking will be addressed after the initial detection system is working.